Supplementary Material: Open-Loop Capacity of the Store

for "Benchmarking Post-Quantum Key Encapsulation in S3-Compatible Object Storage under Client Concurrency"

I. Open-Loop Capacity of the Store

This supplement characterizes the store, not post-quantum cryptography. It is included to check that the key-establishment mode does not move the point at which response time degrades. All numbers are generated by the same pipeline as those of the main paper; experiment labels are those of its mapping table (Open-Loop is batch H of the artifact).

A. Queueing Models

An M/M/c queue has Poisson arrivals at rate λ, c servers and exponential service times with mean S [1]. With $a = \lambda S$ the mean response time is $W = S + C(c,a)\,S/(c-a)$, where C is the Erlang C probability of waiting. Its service time does not depend on load. A *state-dependent* birth–death queue lets the total service rate $\mu(n)$ depend on the number n of requests in the system; its stationary distribution is $p_n \propto \prod_{k\le n} \lambda/\mu(k)$ and its mean response time follows from Little's law.

B. Pre-Registered Queueing Protocol

The 12 open-loop rates are sorted; the 1st, 3rd, … are training points and the 2nd, 4th, … are held out. Models may use training points and the closed-loop runs, and are scored by mean absolute percentage error (MAPE) on the held-out rates at which the system was stable. A rate is stable if every run delivers at least 97 % of the offered rate with a mean queue wait below 10 ms; response times of unstable runs grow with run length and are lower bounds.

C. Results

This subsection describes the store, not post-quantum cryptography, and we include it for one reason: to check that the choice of mode does not move the point of degradation. That closed-loop measurements overstate what an open arrival stream can sustain is established [2]; what we add is where the transition lies for this store.

Two thresholds. We distinguish the *knee*, the interval of offered load in which mean response time first exceeds twice its light-load value, from the *stability boundary*, between the highest rate at which every run kept up and the lowest at which one did not. Under Poisson arrivals of ML-KEM PUTs (Open-Loop), with ρ the offered rate over the peak closed-loop throughput of the same experiment (4,820 and 4,410 requests/s): at 1 KiB the knee and the stability boundary coincide, between 3,600 and 3,900 requests/s ($\rho = 0.75$–0.81); at 64 KiB the knee lies between 3,000 and 3,300 requests/s ($\rho = 0.68$–0.75) and the stability boundary one step higher, between 3,300 and 3,600 ($\rho = 0.75$–0.82). The confirmatory campaign measured ML-KEM only; the exploratory campaign found the knee in the same place for the symmetric-only mode B1 and for P1, which is our evidence that the mode does not move it.

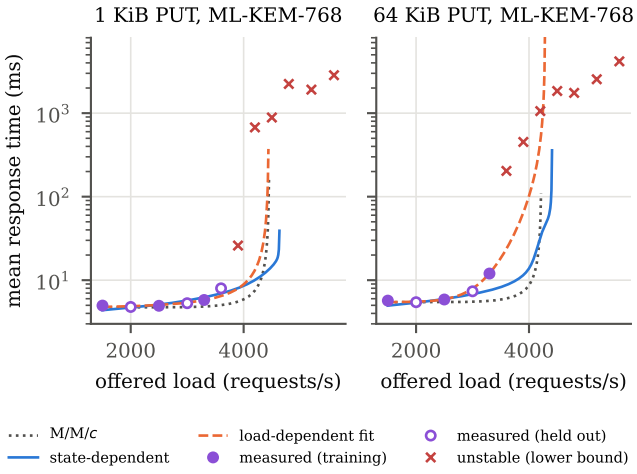


Fig. 1. Open-loop mean response time of ML-KEM PUT against offered load (Open-Loop; log scale), with three models. Filled: training rates; hollow: held-out rates; crosses: unstable rates (lower bounds).

TABLE I
QUEUEING MODELS FOR ML-KEM PUT (REQUESTS/S). STABLE/UNSTABLE: THE STABILITY BOUNDARY, I.E. HIGHEST STABLE AND LOWEST UNSTABLE OFFERED RATE. OBSERVED KNEE: INTERVAL IN WHICH MEAN RESPONSE TIME FIRST EXCEEDS TWICE ITS LIGHT-LOAD VALUE; PREDICTED KNEE: THE RATE AT WHICH EACH MODEL DOES.

Size	Peak	Stable/unstable	Observed knee	Held-out error (MAPE) M/M/c	State	Load	Predicted knee (req/s) M/M/c	State	Load
1 KiB	4,820	3,600/3,900	3,600–3,900	16.3%	7.5%	7.6%	4,300	4,190	4,070
64 KiB	4,410	3,300/3,600	3,000–3,300	12.5%	4.4%	4.5%	4,080	3,890	3,270

Models. We compare three models (Fig. 1, Table I). (a) M/M/c with the light-load service time S_0, capacity μ from the two most overloaded training rates and $c = \mathrm{round}(\mu S_0)$. (b) A state-dependent queue whose service rate $\mu(n)$ is the closed-loop throughput measured with n workers, interpolated linearly; it uses no open-loop data at all. (c) A load-dependent fit $W(\rho) = S_0\,[1 + a\rho^b/(1-\rho)]$ with $\rho = \lambda/\mu$, whose two parameters are fitted to the stable training rates.

On the stable held-out rates (only 3 at 1 KiB and 2 at 64 KiB), M/M/c has a MAPE of 16.3 % and 12.5 %; the state-dependent model 7.5 % and 4.4 %; the load-dependent fit 7.6 % and 4.5 %. The two alternatives are close to each other but not identical. Both roughly halve the error because they let service time rise with load, which M/M/c cannot: the mean service time of a 1 KiB PUT grows from 4.7 ms at the lightest rate to about 51 ms in overload.

The knee is still predicted too late. M/M/c places the knee at 4,300 and 4,080 requests/s and the state-dependent model at 4,190 and 3,890, both above the observed intervals. The fitted model matches at 64 KiB (3,270) and is late at 1 KiB (4,070). The reason is visible in the data: the state-dependent model assumes the store can sustain its mean closed-loop throughput, but in overload the open-loop runs achieved only 4,300 and 4,120 requests/s, below the closed-loop peaks, and closed-loop throughput itself varies by several percent between runs at high concurrency. A system whose capacity fluctuates becomes unstable when the offered rate reaches the lower part of that range, not its mean. With only 3 and 2 stable held-out points per size, these error figures are indicative rather than precise, and the fitted model is a description of this configuration, not a general law.

Summary. For this store, response time doubles at 0.68–0.81 of peak closed-loop throughput and runs become unstable at 0.75–0.82. Letting service time depend on load halves the prediction error of M/M/c below the knee, on very few held-out points; no model we tested predicts the knee from closed-loop capacity.

References

[1] M. Harchol-Balter, *Performance Modeling and Design of Computer Systems: Queueing Theory in Action*. Cambridge University Press, 2013.

[2] B. Schroeder, A. Wierman, and M. Harchol-Balter, "Open versus closed: A cautionary tale," in *Proc. 3rd USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2006, pp. 239–252.